

AN INTELLIGENT SYSTEM FOR DETECTION, TRANSLATION, AND HISTORICAL MAPPING OF ANCIENT INSCRIPTIONS



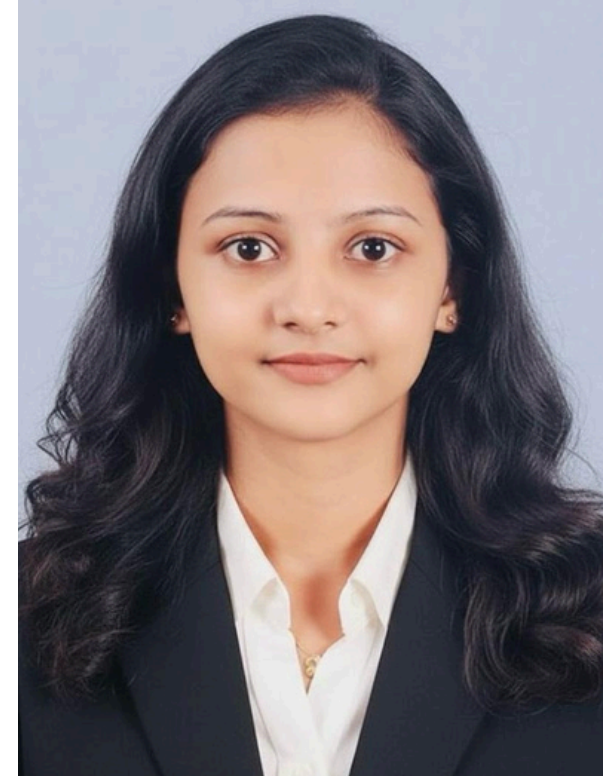
Meet our team



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Why Silent Script ?



- Ancient inscriptions are the key to Sri Lanka's history
- Many are damaged, faded, hard to read.
- Manual transcription is slow & error-prone.
- No proper system for Early Sinhala OCR & translation

Silent Script → AI-powered solution to detect, translate & preserve ✓

AN INTELLIGENT SYSTEM FOR DETECTION, TRANSLATION, AND HISTORICAL MAPPING OF ANCIENT INSCRIPTIONS

Problem Definition

- Researchers/Archaeologists
- Historians/Epigraphists
- Academic Community
- General Public:

Risk Mitigation

- Data Availability
- Model Generalization
- Expert Validation
- Processing Performance

**with overlays
reminding**





Component 1

IMAGE SHARPENING & CHARACTER SEGMENTATION OF ANCIENT INSCRIPTIONS FROM ANURADHAPURA DISTRICT OF SRI LANKA

IT22586070| Handapangoda C.N.

Specialization : Data Science

Problem Definition

- Degraded Image Quality
- Manual Processing
- Character Overlap
- Loss of Historical Data

Proven Gap

- Separate Approaches
- Generic Methods
- Modern Text Focus
- Dataset Gap

Solution Approach

Multi-Stage Enhancement Pipeline

Stage 1: Image Sharpening

Stage 2: Image Preprocessing & Noise Reduction

Stage 3: Intelligent Segmentation

How problems Are Addressed

- Automated Enhancement
- Consistency
- Ancient Script Optimization
- Edge Preservation

User Requirements

- **Archaeologists/Researchers/Historians/Learners**
- Easy upload of inscription images
- High-quality enhanced outputs
- Precise character segmentation
- Easy-to-use interface for learning ancient scripts,
- Clear visualization of segmentation results
- Educational resources and examples

Functional Requirements

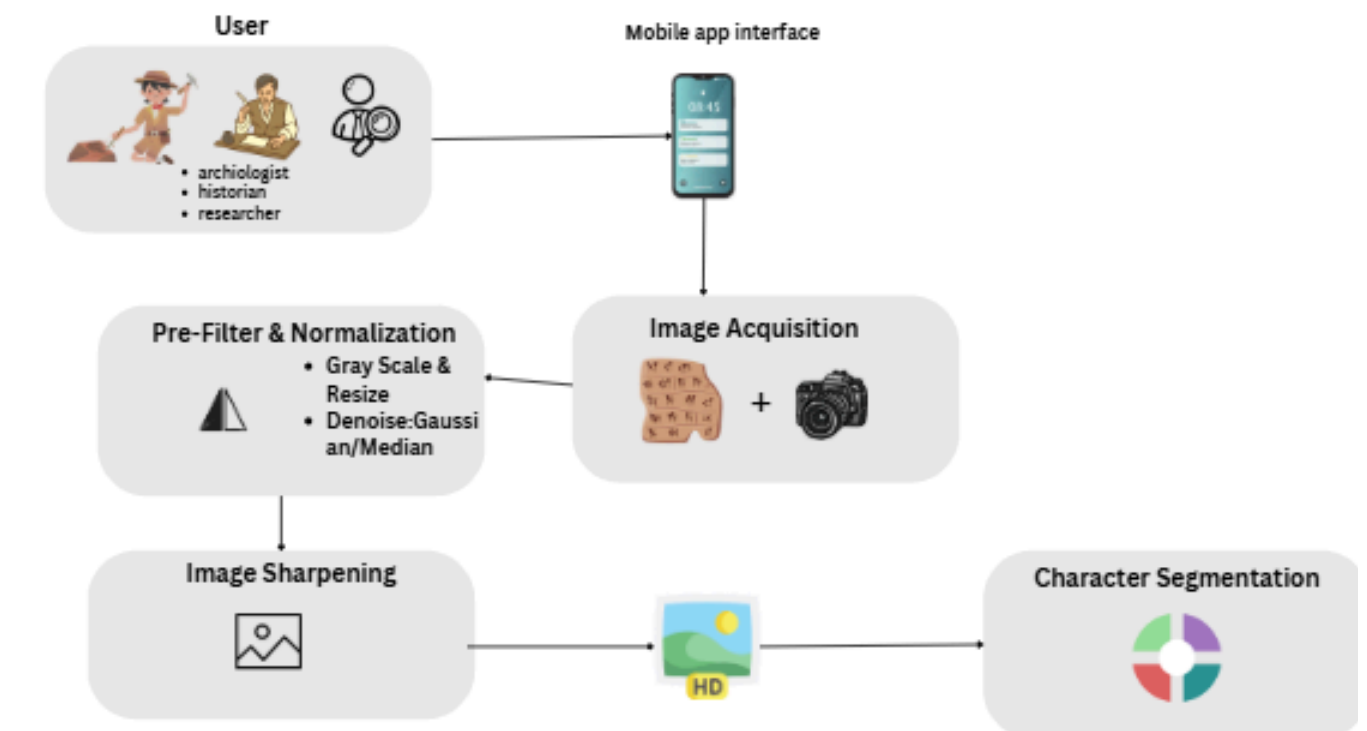
- **Image Upload**
- **Preprocessing**
- **Character Segmentation**
- **Export Options**

Design Excellence

- **3-stage:** Sharpening → Noise Reduction → Segmentation
- **Simple web interface with real-time preview**
- **Specialized for ancient Sri Lankan scripts**

Usser Feedback on Prototype

Will be presented in common





Component 2

CHARACTER RECOGNITION FROM ANCIENT INSCRIPTIONS FROM ANURADHAPURA DISTRICT OF SRI LANKA

IT22089304 | BANDARA K.M.W.G.L.A

Specialization : Data Science

Problem Definition

- OCR fails on Early Sinhala characters.
- Limited training data.
- Manual transcription is slow and inconsistent.

Proven Gap

- OCR works only on modern scripts, not Early Sinhala
- No character-level recognition for ancient inscriptions
- Very limited training data
- Classical methods, no deep learning used
- No confidence scores or uncertainty handling
- Lacks glyph-level OCR capability

Solution Approach

- YOLOv8 for character detection
- Grouping algorithm for word formation
- Confidence scores and uncertainty flags
- Structured output for downstream processing

How problems Are Addressed

- Automated Recognition
- Ancient Script Specialization
- Uncertainty-Aware Pipeline

User Requirements

- **Archaeologists/Researchers/Learners:**
- Automated recognition, confidence scores, uncertainty flags
- Fast digitization, accurate transcription, export capabilities
- Clear visualization, educational access to digitized data

Functional Requirements

- **Image Input**
- **Glyph Detection**
- **Character Recognition**
- **Word Grouping**
- **Confidence Scoring**

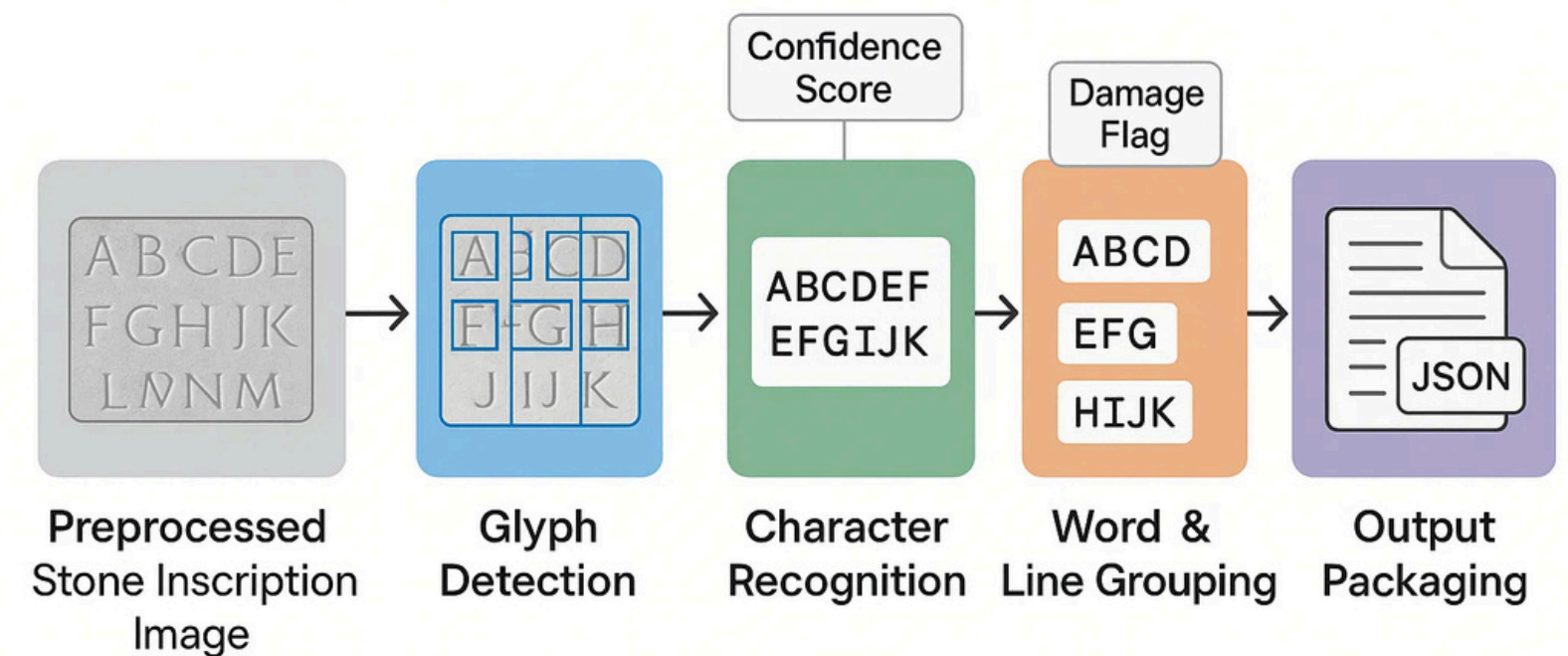
Design Excellence

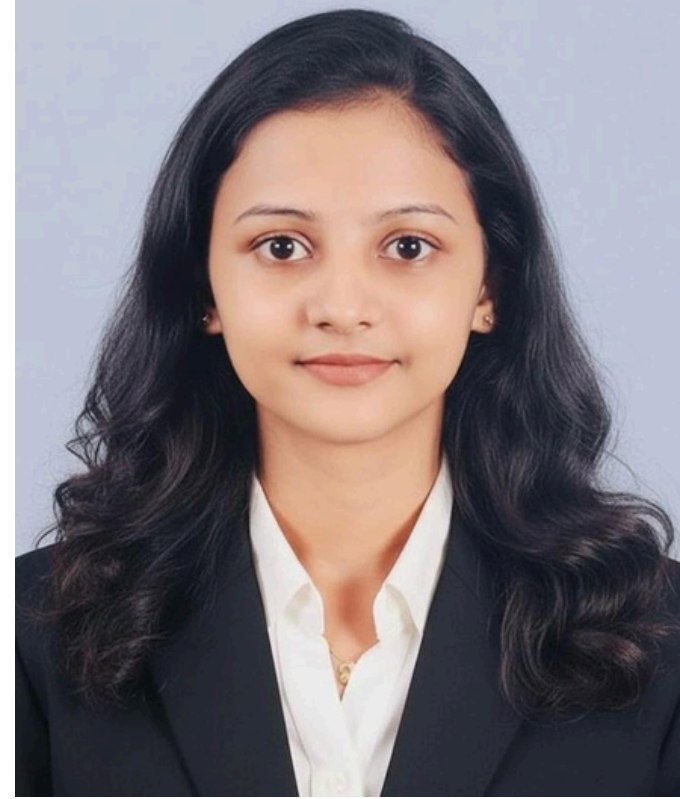
- YOLOv8 + CNN/ViT deep learning pipeline
- Uncertainty-aware with confidence scoring
- Structured output for downstream integration

User Feedback on Prototype

Will be presented in common

Character & Damage Detection Pipeline





Component 3

TRANSLATION AND KNOWLEDGE EXTRACTION OF ANCIENT INSCRIPTIONS FROM ANURADHAPURA DISTRICT OF SRI LANKA

IT22069436 | Hansani B.A.A.

Specialization : Data Science

Problem Definition

- Current systems stop at OCR or partial Sinhala translation
- No Sinhala + English translation with cultural meaning
- Knowledge extraction is missing (Entities & Relations)

Proven Gap

- No complete translation pipeline from Brahmi → Sinhala → English
- Lack of automated systems to extract entities and relationships from inscriptions
- Inscription data remains unstructured and difficult to analyze

Solution Approach

- Transformer-based Brahmi → Sinhala translation
- Brahmi into Sinhala + English Translation together.
- Extract entities & relations from knowledge extraction

How problems Are Addressed

- Full translation pipeline from Brahmi → Sinhala + English
- Use Sinhala as an intermediate language to preserve historical and cultural context
- Apply knowledge extraction techniques to identify key entities and relationships

User Requirements

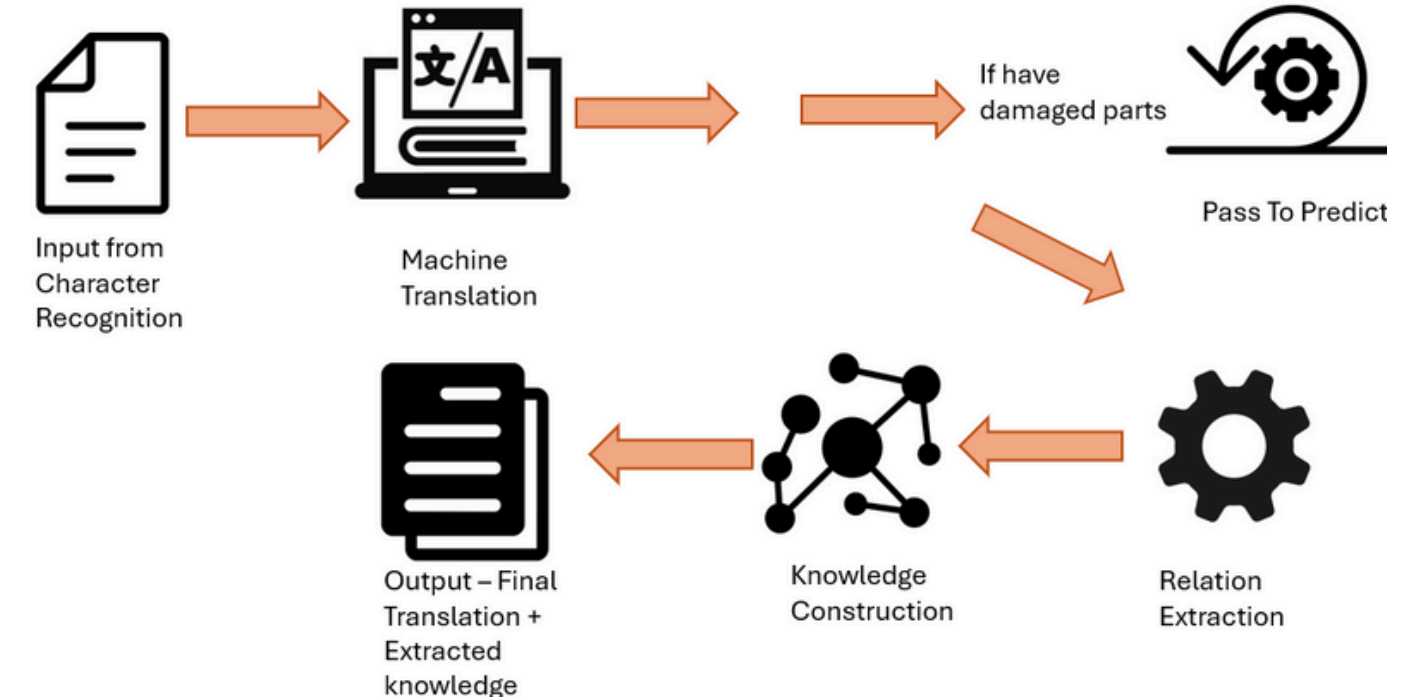
- Researchers/Historians/Learners:: Accurate translations, Brahmi into Sinhala & English.
- Process translations automatically and store extracted knowledge
- Display clear translations and extracted knowledge

Functional Requirements

- Accept digitized Brahmi text
- Translate Brahmi into Sinhala & English
- Extract Key entities like monastries, kingdoms and domains.

Design Excellence/Contribution

- No ready-made dataset and create my own dataset.
- Manually typed and translated all Brahmi inscriptions by myself
- Combines Transformers and mT5 as base model.
- Generative AI and APIs for Sinhala to English Translation.
- NER for Historical knowledge extraction.





Component 4

Multimodal Restoration and Contextual Completion of the Anuradhapura Inscriptions

IT22108890| Mesandu W.M.S.

Specialization : Data Science

Problem Definition

- The project aims to automatically restore damaged characters using deep learning while preserving their original structure

Proven Gap

- No models for Brahmi/Early Sinhala
- Visual and linguistic processing not integrated
- No translation to modern languages
- Synthetic data mismatches weathering patterns
- Text-only methods ignore visual damage

Solution Approach

- I use deep learning based image restoration model trained on synthetically damaged brahmi characters to reconstruct clean inscription images.

How problems Are Addressed

- Realistic synthetic damage is generated to overcome the lack of real damaged data.
- An improved UNet model is trained with composite loss functions to preserve structure and strokes
- Validation metrics are used to select the best performing model.

Training is accelerated using a paid Google colab T4 GPU

User Requirements

- **Historians/Researchers/Learners:**
Upload images, view restorations with confidence scores, access historical parallels, Accurate gap filling, translations

Functional Requirements

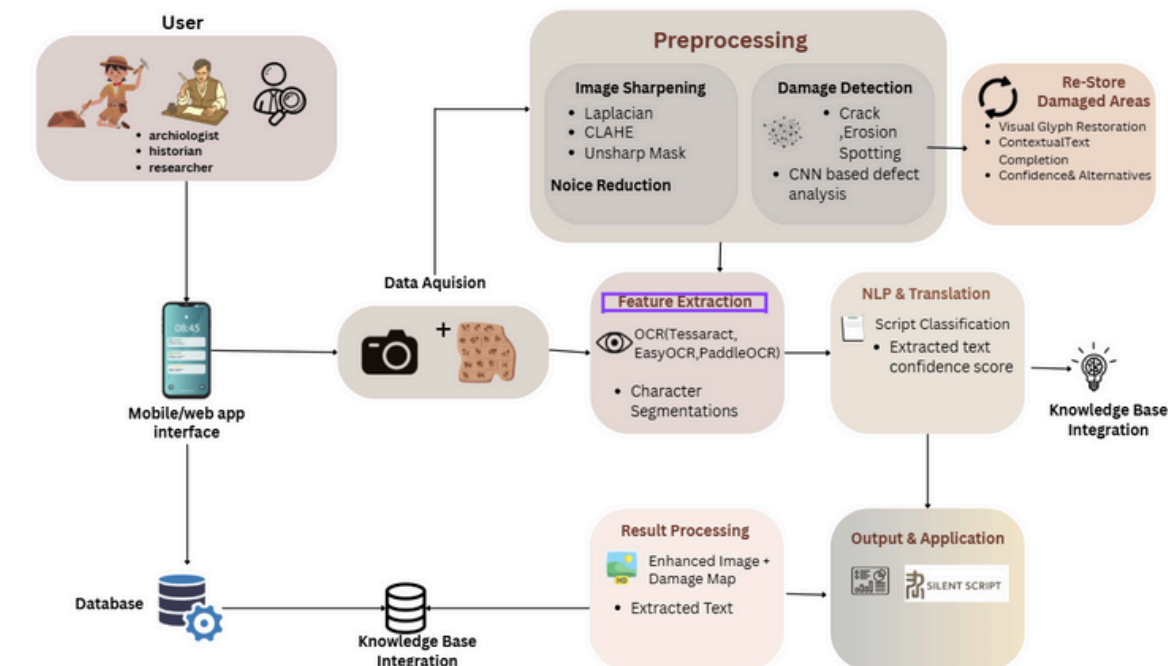
- Image Upload
- Damage Detection
- Visual Restoration
- Textual Gap Filling
- Contextual Suggestions

Design Excellence

- Dual-stream: visual encoder + language encoder
- Multi-task learning with stroke-aware training
- End-to-end restoration and translation pipeline

User Feedback on Prototype

Will be presented in common



Customer Persona

- **Customers:** Archaeological departments, national museums, heritage organizations, university research departments
- **Users:** Archaeologists, epigraphists, historians, heritage conservationists, researchers

Market Size

- **Growing Digital Heritage**
- **Rich Historical Sites**
- **Limited Specialists**
- **High Research Demand**

What Makes It Unique

- A damaged character restoration model trained using deep learning specifically trained for Brahmi letters.
- Trained on efficiently using GPU acceleration with a large dataset

Cost Recovery

- Institutional license: LKR 5,000 – 15,000/year
- Pay-per-use: LKR 50 – 200/inscription
- API subscription: LKR 1,000 – 5,000/month
- Grants/funding: LKR 10,000 – 50,000/project



Thank You